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Shifting soil N regimes over a 70-year chronosequence of wetland reclamation and restoration

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ABSTRACT

Shifting nitrogen (N) cycles are fundamental to the ecosystem functioning during wetland reclamation and restoration. Our study investigated N transformation across a 70-year chronosequence in Northeast China, encompassing natural wetland, cultivated paddy conversed from wetland for approximately 20, 30, 40, 50, and 60 years, and restoration from cultivated cropland for 3, 5, and 10 years. There was a 73.2 % decrease in soil total nitrogen (TN) in the first 40 years of reclamation, with a modest subsequent recovery. After one decade of restoration, soil N content accumulated and increased by 34.7 % from its level after 60-year reclamation. The inorganic N concentrations along the chronosequence echoed this pattern. Mineralization primarily drove increased nitrous oxide (N_2O) production potential during the early (20–30 years) and later (40–60 years) reclamation phases. Conversely, the net nitrification rate significantly impacted N_2O production during restoration, with a coefficient of 0.43 ($p < 0.05$). In pristine and restored wetlands, nitrification positively correlated with nutrient content, including total carbon (TC) and TN. These dynamic shifts in N cycling, particularly regarding N_2O emissions during the wetland-cropland transition, highlighted the need for tailored management strategies at different reclamation and restoration phases. Our detailed assessment of N fluxes and pool sizes throughout the chronosequence enriches the understanding of N dynamics and could be used as a benchmark for validating ecosystem models for simulating long-term N dynamics during land use change.

1. Introduction

Wetlands are crucial for mitigating climate change by serving as significant carbon pools and facilitating nutrient removal and water purification (Choudhury et al., 2018). They are particularly adept at nutrient retention, transforming these elements into plant biomass, with nitrogen (N) being a critical nutrient that wetlands sequester (Laine et al., 2018; Singh, 2021; Zhang et al., 2019). Nonetheless, more than half of the natural wetlands in regions such as North America, Europe, Australia, and China have been transformed into farmland (Davidson, 2014; Fluet-Chouinard et al., 2023; Mitsch and Gosselink, 2007; Pang et al., 2020). The conversion of wetlands into croplands profoundly impacts chemical, physical, and biological properties in soils (Zhu et al., 2022), which can influence agricultural productivity and biogeochemical cycles (Wan et al., 2021). Intensified farming practices have been linked to substantial soil N depletion, with losses ranging from 25 % to 60 % (Sun et al., 2015; Weier, 1994). Conversely, restoring wetlands from agricultural lands has shown promise in reviving N levels and enhancing overall ecosystem function (Temmink et al., 2022; Wan et al., 2021; Zhong et al., 2021). For the welfare of humanity and sustainable development, wetland restoration projects aimed at reinstating their structure and ecological roles have been initiated since the late 20th

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century (Perez-Brandan et al., 2019).

Various N transformation processes govern the fates of soil N during wetland cultivation and restoration (Jiang et al., 2021; Laine et al., 2018), such as N mineralization (Balota et al., 2004; Borken and Matzner, 2009), ammonification (Jiang et al., 2021; Risch et al., 2019), nitrification (Martens-Habbena et al., 2009; Mohamed et al., 2010), and denitrification (Hartzog et al., 2017; Starr et al., 2022). Studies have indicated that prolonged cultivation diminishes soil mineralization and microbial N immobilization (Li and Lang, 2014), potentially undermining ecosystem productivity and resilience (Liu et al., 2021; Reich et al., 1997). Furthermore, it has been suggested that reclamation can induce a priming effect that escalates N loss through N_2O production (Daly and Hernandez-Ramirez, 2020; Xu et al., 2023). In addition, reclamation activities that stimulate nitrification can increase substrate availability for denitrification, which is a primary route for N_2O production (Liang et al., 2022; Mørkved et al., 2007; Xu et al., 2023). These studies have elucidated N dynamics following land use changes in wetlands. However, the understanding of N cycling across long chronosequence is still limited.

Space-for-time (SFT) substitution offers a robust approach for examining slow-evolving biogeochemical processes that are impractical to monitor directly over brief spans (Pickett, 1989; Wogan and Wang, 2018). This method assumes that spatial variance in land characteristics can serve as a proxy for their temporal progression (Doetterl et al., 2018). In fields of ecology and evolution, SFT substitution is utilized to investigate long-term vegetative succession and nutrient cycling (Buma et al., 2017; Johnson and Miyanishi, 2008). By comparing sites across different stages of reclamation, researchers can deduce the progression of N transformation over long periods. For example, one study reported an increase in soil total N (TN) from 0.51 g/kg to between 1.60 g/kg and 1.53 g/kg following 1–3 years of reclamation (Xu et al., 2020). In contrast, a decrease in TN from 1.09 g/kg to 0.98 g/kg was documented over 32 to 40 years of coastal wetlands cultivation (Xie et al., 2020). In the natural restoration of former agricultural lands, soil microbes were initially constrained by total carbon (TC) and TN in the early successional stages (3–52 years), shifting to P limitations in later stages (80–120 years) (Zhou et al., 2023). Nonetheless, due to gaps in our comprehensive understanding of long-term N dynamics, pinpointing the definitive limiting factors of N dynamics throughout the full spectrum of wetland reclamation and restoration remains challenging (Karki et al., 2021).

In this study, we employed a combined approach of field experimentation and *meta*-analysis to explore shifts in the nitrogen (N) regime throughout a 70-year continuum of wetland reclamation and restoration. The study's aims were two-fold: (i) to quantify the changes in soil N—both in terms of content and turnover rates—across the progression of wetland reclamation to restoration, and (ii) to examine the relationships among crucial N processes, such as mineralization, ammonification, nitrification, and denitrification, in conjunction with soil nutrients, moisture, pH levels, and microbial biomass over this extended period. Our hypotheses posited that (i) soil N contents and turnover rates would diminish during reclamation but would experience a resurgence during restoration, and (ii) the trajectory of soil N would be significantly influenced by the interplay between nitrification and organic N mineralization throughout the stages.

2. Materials and methods

2.1. A global synthesis

A global synthesis was conducted to realize the comprehensive results in change of N contents (including TN, MBN, NH_4^+-N, and NO_3^--N) along the chronosequence of wetland reclamation and restoration. Global data on N contents in wetlands and croplands were obtained from peer-reviewed articles published between 1982 and 2023. The publication databases, Web of Science, Google Scholar, and the Chinese National Knowledge Infrastructure were searched using the following keywords: (carbon* OR nitrogen*) AND (marsh* OR mangrove* OR freshwater wetland* OR riparian wetland* OR peatland* OR fen* OR bog) AND (land use* OR drained wetland* OR agriculture* OR restored* OR pasture*). Publications were screened if 1) the study included natural wetlands and croplands. 2) N contents in different forms (TN, MBN, NH_4^+-N, and NO_3^--N) were reported for wetlands and croplands. Finally, a total of 818 observations from 84 publications were included in the study (Table S4). Available auxiliary information on the sampling sites, such as latitude, longitude, mean annual air temperature, mean annual precipitation, soil moisture, and soil pH, was also retrieved. To verify the global findings at the scale of site and further detect the influence of environmental factors on soil N cycle along the chronosequence of wetland reclamation and restoration, we conducted a field study in Sanjiang Plain.

2.2. Site description and sampling

We conducted a field study on the Sanjiang Plain (47°35′ N, 133°31′ E), northeastern China (Fig. 1). The sampling sites spanned three distinct land-use types: natural wetlands, reclaimed agricultural lands with paddy fields, and wetlands undergoing restoration. The undisturbed wetland, part of the Sanjiang Nature Reserve, features vegetation cover predominantly composed of *Carex lasiocarpa*, *Carex pseudocuraica*, and *Carex meyeriana*. The agricultural lands converted from natural wetlands have been under rice cultivation, with rice typically sown in May and harvested in September. Chemical fertilization in these paddies is conservative, including urea and diammonium phosphate, at a rate of 0.05 t/ha (Yang et al., 2018), far below the global average of 0.74 t/ha (Lu and Tian, 2017). Restored wetlands have been converted from abandoned croplands. After abandonment, the restored wetlands were never cultivated or mowed.

We used the Space-for-time (SFT) substitution to explore the chronosequence of land use change, a commonly applied approach to evaluate long-term processes (Walcker et al., 2018). The chronosequence includes one natural wetland site and reclaimed sites converted from natural wetlands for about 20, 30, 40, 50, and 60 years, and restored wetland sites of 3, 5, and 10 years, respectively. Except for the differences in land-use type and restoration time, the selected sampling sites had the same geographical characteristics, including elevation, slope gradient, and parent soil material. The range of elevation is 50–60 m above sea level, and the slope gradient is between 0.01 and 0.02 %. The parent materials are quaternary alluvial sediments (Lin et al., 2015). At each land-use site, we designated three random plots (10 m × 10 m) for soil sampling. After harvest in October 2021, we collected three soil samples from each plot of natural wetlands, croplands, and restored wetlands. We partitioned every sample into ten soil depths: 0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, 60–70 cm, 70–80 cm, 80–90 cm, and 90–100 cm in natural wetlands and restored wetlands. Due to the plowing activity occurring up to 20 cm deep post-harvest, we adjusted the soil core divisions accordingly for the farmlands. Every soil core was divided into nine soil depths: 0–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, 60–70 cm, 70–80 cm, 80–90 cm, and 90–100 cm. We meticulously removed stones and rhizomes to prepare a composite sample for each soil depth. In total, our collection included 255 soil samples.

Fresh soils (300 g) from each sample were promptly transported to our laboratory within 24 h and preserved at 4 °C for subsequent analysis of soil water content (SWC), nitrite (NO_2^--N), nitrate (NO_3^--N), ammonium (NH_4^+-N), microbial biomass carbon (MBC), microbial biomass nitrogen (MBN), microbial biomass phosphorus (MBP), and microbial biomass sulfur (MBS). Additionally, we prepared air-dried subsamples (100 g) for evaluating soil pH, TC, TN, total phosphorus (TP), and total sulfur (TS).

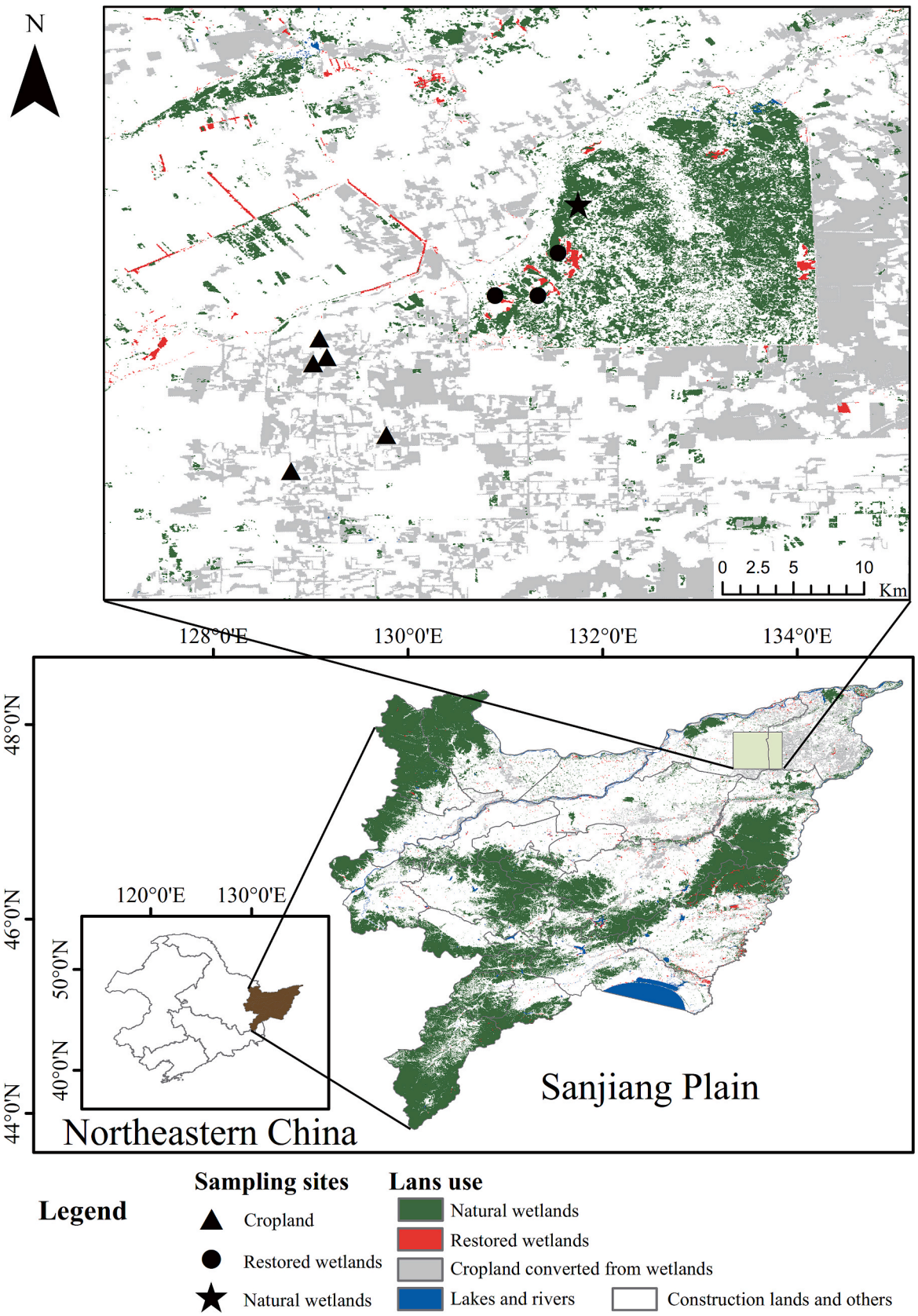


Fig. 1. Schematic map of sampling sites in the Sanjiang Plain.

2.3. Soil physicochemical characteristic analyses

The soil temperature (ST) was recorded at each collection site using a T-105 field thermometer (Invisible, China). For TC and TN quantification, soil samples were pulverized and dried at 65 °C for 48 h before analysis with a Vario Macro Cube elemental analyzer (Elementar Company, Germany). TP and TS were determined following acid digestion with HF-$HClO_4$ and measured using ICP-AES (Shimadzu, ICPS-7500, Japan). The soil pH was assessed with a PHS-25 pH meter (Shanghai INESA Scientific Instrument Co. Ltd, China), utilizing a 1:10 soil-to-water ratio (An et al., 2022b).

The SWC was determined by desiccating 5 g of fresh soil at a constant temperature of 105–108 °C until a stable mass was achieved, enabling the calculation of water loss relative to that of the dried soil (Mo et al., 2020). To quantify NO_2^--N, NO_3^--N and NH_4^+-N, we used 5 g of fresh soil to 50 mL of 2 M KCl for extraction. The solution was agitated for one hour and then filtered; the filtrate was analyzed using a continuous flow analytical system (San++ system, Skalar, Holland) to determine NO_2^--N, NO_3^--N, and NH_4^+-N (An et al., 2022a). MBC and MBN were quantified by measuring the C and N extracted with 0.5 mol/L K_2SO_4 from fresh soils fumigated with $CHCl_3$ and comparing them to those extracted from nonfumigated soils (Witt et al., 2000). MBP was determined by the difference in the amount of P extracted with 0.5 mol/L $NaHCO_3$ (pH 8.5) from chloroform-fumigated versus nonfumigated fresh soil. Similarly, MBS was assessed by the difference in the amount of S extracted with 0.01 mol/L $CaCl_2$ from chloroform-fumigated and unfumigated fresh soils. All parameters were weighted by dry soil.

2.4. Soil incubation experiment

For each soil sample, >50 g of fresh soil was preincubated in the dark at 25 °C for 48 h (Frasier et al., 2010). Following homogenization, the preincubated soil (50 g) was placed in a 520 mL serum bottle and wetted with 100 mL sterile distilled-deionized water (ddH_2O) (Ding et al., 2010). The flasks were then sealed and purged by evacuating the air and filling them with pure N_2 (99.999 % purity) to guarantee anoxic conditions (Jia et al., 2006). Subsequently, the samples were incubated in the dark at 25 °C for 6 h (Yu et al., 2001). The extracted gas samples were analyzed for N_2O concentration on a gas chromatograph (GC; Agilent 7820A, USA) equipped with a flame ionization detector (FID), and the N_2O production potential (PP N_2O) was calculated as (Yang et al., 2022):

$$\mathrm{PP\,N_2O} = \frac{(\mathrm{C} - \mathrm{C_0}) \times (\mathrm{V} - 100) \times 273 \times 44}{1000 \times 22.4 \times (273 + \mathrm{T}) \times \mathrm{m} \times \mathrm{t}}$$

where PP N_2O is the soil N_2O production potential [μg N_2O kg^{-1} (dry weight) h^{-1}]; C is the N_2O concentration at the end of incubation [mg/L]; C_0 is the N_2O concentration at the beginning of incubation [mg/L]; V is the volume of containers [mL]; T is the incubation temperature [°C]; m is the dry soil weight in each bottle [mg]; and t is the time of incubation [h]; 44 is the molecular mass of N_2O.

The rate of net ammonification (R_a) during the incubation period was calculated from the difference between the initial and final concentrations of NH_4^+-N as follows (Liu et al., 2021; Wang et al., 2017):

$$\mathrm{R_a} = \frac{\mathrm{NH_4^+} - \mathrm{N_{T1}} - \mathrm{NH_4^+} - \mathrm{N_{T0}}}{\mathrm{T_1} - \mathrm{T_0}}$$

where NH_4^+-N_{T0} is the initial concentration of NH_4^+-N and NH_4^+-N_{T1} is the concentration of NH_4^+-N at the end of the incubation.

The rate of net nitrification (R_n) during the incubation period was calculated from the difference between the initial and final concentrations of NO_3^--N as follows (Robertson et al., 1999; Shi et al., 2016):

$$\mathrm{R_n} = \frac{\mathrm{NO_3^-} - \mathrm{N_{T1}} - \mathrm{NO_3^-} - \mathrm{N_{T0}}}{\mathrm{T_1} - \mathrm{T_0}}$$

where NO_3^--N_{T0} is the initial concentration of NO_3^--N, and NO_3^--N_{T1} is the concentration of NO_3^--N at the end of the incubation.

The rate of net mineralization (R_m) during the incubation period was calculated from the difference between the initial and final concentrations of mineral N (NH_4^+ + NO_3^-) with the following equation (Wang et al., 2017):

$$R_m = R_a + R_n$$

where R_a is the rate of net ammonification and R_n is the net nitrification rate.

The nitrate in the mineral N in the incubated sample (R_{an}) was calculated as follows (Nadelhoffer et al., 1984; Recous et al., 1990; Yan et al., 2018):

$$\mathrm{R_{an}} = \frac{\mathrm{NO_3^-} - \mathrm{N_{T1}}}{\mathrm{NO_3^-} - \mathrm{N_{T1}} + \mathrm{NH_4^+} - \mathrm{N_{T1}}}$$

where NO_3^--N_{T1} is the concentration of NO_3^--N at the end of the incubation, and NH_4^+-N_{T1} is the concentration of NH_4^+-N at the end of the incubation.

2.5. Auxiliary data

In this study, a global land cover map was obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) land cover data product (MCD12Q1) with a spatial resolution of 500 m in 2012 (https://modis.gsfc.nasa.gov/). Datasets for wetland distribution, wetlands reclaimed to croplands, and wetlands restored from croplands during 1990–2015 on the Sanjiang Plain (47°35′ N, 133°31′ E) were provided by the "Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences-Northeast Asia Resource and Environment Big Data Center (https://www.igadc.cn/)".

2.6. Statistical analysis

In global synthesis and field experiment, the soil N content and cycling rates were log-transformed before regression analysis to meet the assumptions of normality and homoscedasticity. Polynomial regression analysis was used to examine trends across the chronosequence of reclamation and restoration. We used non-linear least squares to fit the polynomial curves, guided by the coefficient of determination (R^2) and the Akaike information criterion (AIC) for model selection. In our field experiment, we applied the Wilcoxon test to assess differences in soil N dynamics between pristine wetlands, croplands, and restored wetlands, with subsequent *p*-value adjustments. All the statistical procedures were performed in R (Version 4.1.1). For polynomial regressions, the 'ggpmisc' package was used. We utilized Pearson's correlation analysis, facilitated by the "corrplot" package in R, to probe the relationships between soil attributes, N transformations, and PP N_2O. Due to the important influence of substrates and microbes on the N cycle (Kuypers et al., 2018), we divided environmental factors into two groups: elements and biomass. Elements include TC, TN, TP, and TS, and biomass includes MBC, MBN, MBP, and MBS. The influence of environmental factors and N transformation rates on PP N_2O was further scrutinized through random forest (RF) analysis, employing the 'rfPermute' package to evaluate the significance of the model and its predictors (Jiao et al., 2018). This analysis assessed predictor importance by observing the increase in mean square error (MSE) when a predictor is omitted. We also conducted structural equation modeling (SEM) using Amos software (Version 22). We used the SEM approach to evaluate how soil inorganic N and N transformation rates were influenced by environmental factors along the wetland reclamation and restoration chronosequence. In the model, we tested the influences of reclamation and restoration years, soil depth, soil chemical properties (TC, TN, TP, and TS), and soil microbial properties (MBC, MBN, MBP, and MBS), and soil available N contents (NH_4^+-N and NO_3^--N) on soil N cycling rates (R_m, R_n,

and PP N_2O). The fit of our SEM was gauged using several indices: the chi-square to degrees of freedom ratio (χ^2/df), the Comparative Fit Index (CFI), and the Normed Fit Index (NFI). A lower χ^2/df and root mean square error of approximation (RMSEA), coupled with CFI values exceeding 0.95 and high NFI values, indicated a robust model fit.

3. Results

3.1. Global synthesis of soil N contents during wetland reclamation and restoration

Our global synthesis revealed a decline in soil N content following the transition of wetlands to croplands during the initial 1–11 years of reclamation (as detailed in Table S3). In contrast, wetland restorations led to an increase in soil N content (depicted in Fig. 2). Specifically, during three decades of agricultural use, the mean value of soil TN levels decreased by 54.80 % (Table S3), and the median value of soil TN levels decreased by 63.44 % (Table S4). After 16 years of restoration, however, the mean value of TN rebounded only 6.43 % (Table S3). After 70 years of cultivation, the soil MBN was reduced but later increased again (Fig. 2c). Notably, even after a century of agricultural use, the mean value of soil MBN exhibited a minimal reduction of 0.12 % (Table S3), and the median value of soil MBN exhibited a reduction of 44.58 % (Table S4). In stark contrast, the mean value of soil MBN significantly increased by 134.42 % after 55 years of restoration (Table S3), and the median value of soil MBN increased by 112.07 % after 50 years of restoration (Table S4). Additionally, the concentrations of the soil inorganic N, NH_4^+-N and NO_3^--N, increased under the influence of wetland restoration from previously cultivated lands (Fig. 2d & 2e).

3.2. Soil N dynamics along the reclamation and restoration chronosequence

Our study revealed significant alterations in the soil N content throughout a 70-year span of wetland reclamation and restoration ($p < 0.05$, Fig. S2 and Fig. 3). Soil TN experienced a pronounced decline through the reclamation timeline, plummeting by 73.19 % after four decades (Fig. 3a and Table 1). This decline was followed by a modest resurgence in the subsequent 50th and 60th years (Fig. 3a). After restoration, TN levels fluctuated, registering approximately 1.29 ±

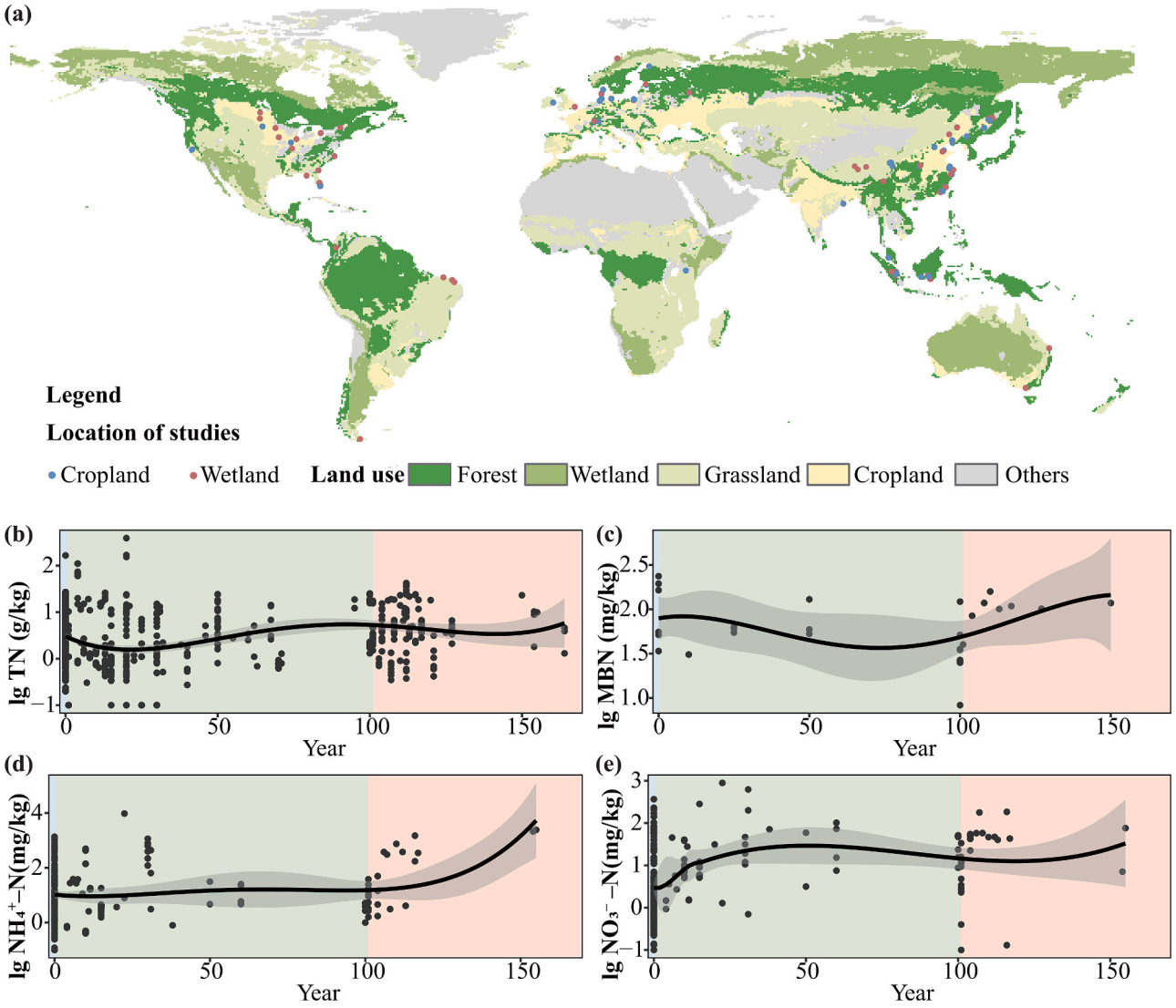


Fig. 2. Synthesized soil nitrogen content across the chronosequence of wetland reclamation and restoration. (a) Study locations of wetlands and croplands in the global *meta*-analysis. (b-e) Changes in soil total nitrogen (TN; b), microbial biomass nitrogen (MBN; c), ammonium (NH_4^+-N; d), and nitrate (NO_3^--N; e) contents during wetland reclamation and restoration are also shown by polynomial fitted curves, whose bands represent 95 % confidence intervals. The background color in each panel represents the different land-use types: blue indicates wetlands, green indicates croplands, and red background indicates restored wetlands. Notably, complete wetland restoration occurs after being cultivated for 100 years.

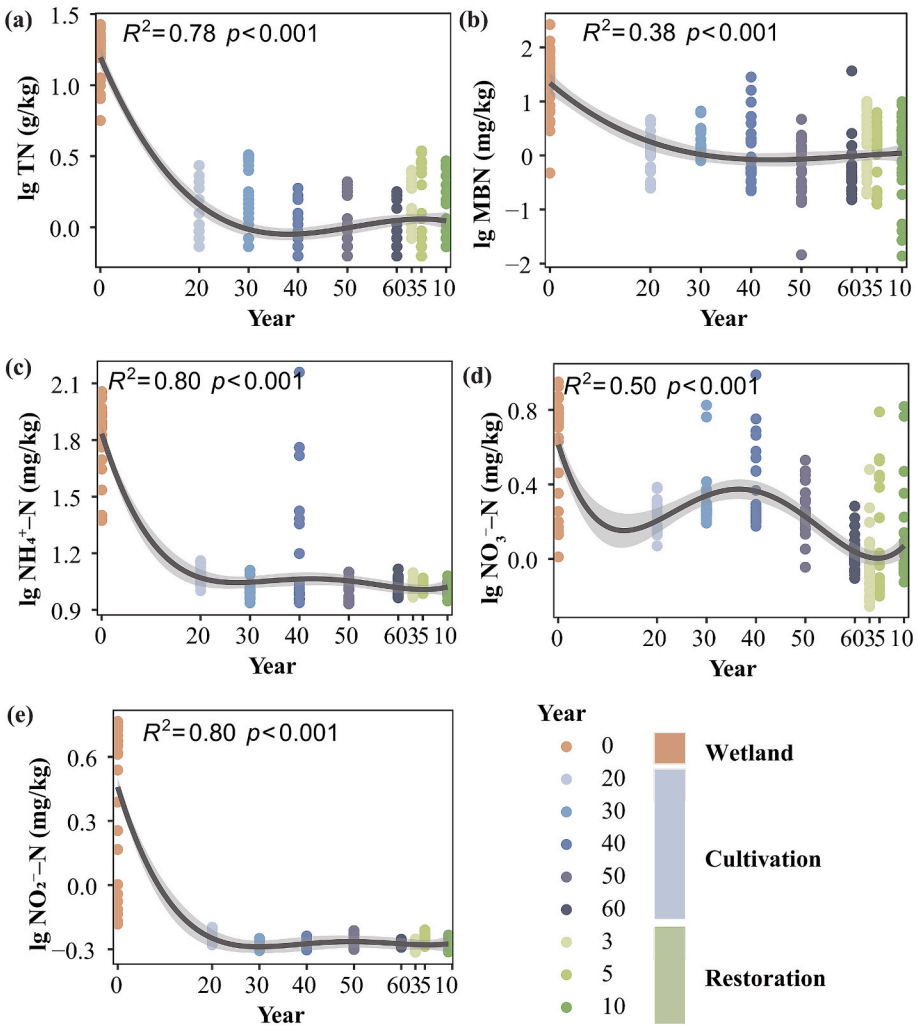


Fig. 3. Changes in soil nitrogen contents the chronosequence of wetland reclamation and restoration. (a) total nitrogen (TN), (b) microbial biomass nitrogen (MBN), (c) ammonium (NH_4^+-N), (d) nitrate (NO_3^--N), and (e) nitrite (NO_2^--N) contents in different years of wetland reclamation and restoration. Polynomial fitted curves are given, and the bands represent 95 % confidence intervals. The color in each point represents the different land-use types.

0.10g kg^{-1} after 3 years and 1.37 ± 0.17g kg^{-1} after 5 years, without displaying a definitive temporal pattern. The MBN in the cropland decreased within the initial 40 years of cultivation and then plateaued (Fig. 3b). In natural wetlands, the NH_4^+-N concentrations were the highest, ranging from 23.62 mg kg^{-1} to 113.95 mg kg^{-1}. However, these levels decreased more than fivefold during the 30-year cultivation period (Fig. 3c). Interestingly, the NH_4^+-N content increased again between the 40th and 60th years of reclamation. A similar pattern was observed for soil NO_2^--N. The soil NO_3^--N concentration was initially highest under undisturbed wetland conditions (4.95 ± 0.43 mg N kg^{-1}) but decreased as the reclamation process processed, decreasing to its lowest value after 3 years of restoration.

Substantial variations in the soil N content across the reclamation-restoration chronosequence were evident at all the measured soil depths (Fig. S3). In natural wetlands, soil TN was substantially greater, nearly tenfold, than in croplands and restored wetlands. A gradient of TN decline was noted with increasing soil depth, which was the most pronounced in natural wetlands. The soil MBN in wetlands was up to 40 times greater than in cropland and restored wetland counterparts (Table 1). While MBN levels in croplands and restored wetlands did not exhibit a consistent pattern across the soil profile, concentrations were notably higher in the upper soil strata (0–50 cm) and modestly reduced at greater depths (50–100 cm). The soil NO_3^--N concentration peaked in natural wetlands, with restored wetlands following suit. This pattern persisted consistently throughout all soil depths, as shown in Fig. S3.

3.3. Potential rates of soil N processes

We observed significant fluctuations in the soil N process rates across different soil types throughout the reclamation and restoration chronosequence (Fig. S4 and Fig. 4). The PP N_2O value was notably higher in the natural wetland (3.72 ± 0.35μg $kg^{-1}h^{-1}$) than in the 30-year reclaimed cropland (0.17 ± 0.03 μg $kg^{-1}h^{-1}$). This value showed an upward trend after 40 years of reclamation (Fig. 4). In the restored wetland, PP N_2O increased to 1.93 ± 0.93 μg $kg^{-1}h^{-1}$ following five years of restoration (Table 1). The rate of R_n sharply decreased from 2.56 mg $kg^{-1}h^{-1}$ in pristine wetlands to 0.15 mg $kg^{-1}h^{-1}$ in croplands reclaimed for both 30 and 40 years, but it increased to 0.42 mg $kg^{-1}h^{-1}$ in wetlands restored for 10 years (Table 1). Conversely, the rate of R_{an} initially increased during the first two decades of reclamation before decreasing. At the end of the reclamation chronosequence, the R_n rose from 0.35 % after 60 years of reclamation to 0.44 % after 10 years of recovery (Table 1). The rate of R_m had significant differences between croplands and restored wetlands ($p < 0.05$; Fig. S4), yet it did not display a clear temporal trend from reclamation to restoration. The rate of R_a underwent a gradual increase in the initial 20 years of reclamation and then leveled off, with no significant differences detected across wetlands, croplands, or restored wetlands (Fig. S4).

Table 1
Soil N dynamics (mean ± se) in different years of wetland reclamation and restoration.

Land use	Year	TN (g kg^{-1})	MBN (mg kg^{-1})	NH_4^+-N (mg kg^{-1})	NO_3^--N (mg kg^{-1})	NO_2^--N (mg kg^{-1})	N_2O (μg $kg^{-1}h^{-1}$)	R_m (mg $kg^{-1}h^{-1}$)	R_a (mg $kg^{-1}h^{-1}$)	R_n (mg $kg^{-1}h^{-1}$)	R_{an} (%)
Wetland	0	17.77 ± 1.05^{a}	44.28 ± 9.79^{a}	73.82 ± 4.69^{a}	4.95 ± 0.43^{a}	3.72 ± 0.35^{a}	2.4 ± 0.66^{d}	0.2 ± 1.56^{d}	−2.36 ± 1.59^{f}	2.56 ± 0.31^{f}	0.27 ± 0.03^{f}
Cropland	20	1.17 ± 0.12^{c}	1.46 ± 0.2^{bc}	12.06 ± 0.21^{bd}	1.69 ± 0.06^{c}	0.56 ± 0^{b}	0.24 ± 0.02^{ad}	−1.25 ± 0.04^{c}	−1.42 ± 0.03^{de}	0.18 ± 0.01^{d}	0.44 ± 0.01^{a}
	30	1.55 ± 0.16^{b}	1.89 ± 0.28^{bc}	10.16 ± 0.17^{b}	2.22 ± 0.23^{b}	0.53 ± 0^{d}	0.17 ± 0.03^{c}	−0.93 ± 0.03^{b}	−1.08 ± 0.02^{cdef}	0.15 ± 0.02^{e}	0.45 ± 0.01^{a}
	40	0.96 ± 0.07^{bc}	3.21 ± 1.17^{b}	19.79 ± 5.36^{b}	2.64 ± 0.35^{b}	0.53 ± 0^{d}	0.18 ± 0.04^{c}	−0.9 ± 0.51^{abcd}	−1.06 ± 0.51^{ac}	0.15 ± 0.02^{de}	0.36 ± 0.03^{ce}
	50	1.04 ± 0.09^{bc}	0.97 ± 0.19^{d}	10.02 ± 0.26^{b}	1.91 ± 0.11^{bc}	0.54 ± 0.01^{c}	0.23 ± 0.04^{bc}	−0.54 ± 0.11^{a}	−0.86 ± 0.06^{ac}	0.33 ± 0.05^{bc}	0.44 ± 0.01^{a}
	60	0.98 ± 0.07^{bc}	1.92 ± 1.34^{bd}	10.53 ± 0.18^{d}	1.09 ± 0.05^{e}	0.54 ± 0^{c}	0.31 ± 0.09^{abc}	−0.17 ± 0.26^{a}	−0.43 ± 0.27^{ab}	0.26 ± 0.03^{c}	0.35 ± 0.02^{de}
Restored wetland	3	1.29 ± 0.1^{b}	3.03 ± 0.52^{bd}	10.26 ± 0.13^{b}	0.98 ± 0.09^{f}	0.51 ± 0^{e}	0.42 ± 0.13^{ab}	−0.73 ± 0.06^{ab}	−1.05 ± 0.02^{be}	0.32 ± 0.05^{bc}	0.39 ± 0.02^{bc}
	5	1.37 ± 0.17^{c}	1.96 ± 0.27^{bc}	10.77 ± 0.1^{b}	1.38 ± 0.22^{de}	0.54 ± 0^{d}	1.93 ± 0.93^{e}	−0.76 ± 0.03^{a}	−1.1 ± 0.02^{ac}	0.34 ± 0.02^{a}	0.43 ± 0.01^{ab}
	10	1.32 ± 0.14^{bc}	2.25 ± 0.43^{bc}	10.15 ± 0.11^{c}	1.28 ± 0.18^{df}	0.53 ± 0.01^{c}	1.28 ± 0.75^{ab}	−0.55 ± 0.07^{ab}	−0.98 ± 0.02^{d}	0.42 ± 0.06^{ab}	0.44 ± 0.02^{abd}

The different letters in one column indicate significant differences at the $p = 0.05$ level. Year: the reclamation or restoration year; TN: total nitrogen; MBN: microbial biomass nitrogen; NH_4^+-N: ammonium; NO_3^--N: nitrate; NO_2^--N: nitrite; R_m: net N mineralization rate; R_a: net ammonification rates, R_n: net nitrification rates, R_{an}: nitrate in mineral N in the incubated sample, and N_2O: N_2O production potential.

Our analysis highlighted distinct variations in soil N cycling rates along the depth profiles of pristine wetlands, croplands, and restored wetlands (Fig. S5). The rate of PP N_2O decreased with increasing soil depth, a pattern that was more pronounced in the restored wetland than in the natural wetlands and croplands (Fig. S5). The highest levels of soil PP N_2O were found in natural wetlands, followed closely by restored wetlands (Fig. S4), and this pattern held consistent across all soil depths. The rate of R_n decreased, similar to that of PP N_2O, which was the most substantial in wetlands and decreased with soil depth, especially in the restored wetlands. The R_a mirrored the trend observed in the rate of R_m across all soil depths and throughout the reclamation and restoration sequence, with the most significant fluctuations occurring in pristine wetlands (Fig. S5).

3.4. Relationships between soil N dynamics and environmental factors

Our study revealed dynamic shifts in the environmental factors affecting PP N_2O across the reclamation and restoration sequence (Fig. 5). In pristine wetlands, we found a strong positive correlation between PP N_2O and SWC, R_n, R_{an}, NO_2^-, NO_3^-, NH_4^+, MBC, and TN ($p < 0.01$; Fig. 5a). In the early stages of wetland reclamation (20 and 30 years), PP N_2O was significantly positively correlated with SWC, R_n, R_{an}, NO_2^--N, NH_4^+-N, MBC, MBN, MBS, and TP ($p < 0.01$; Fig. 5b). After 40–60 years of agricultural use, the correlations of PP N_2O shifted to significantly positively correlated with SWC, R_m, R_a, NO_2^--N, NH_4^+, MBC, MBS, TN, TC, TP, and TS ($p < 0.05$; Fig. 5c). Post restoration, PP N_2O was significantly positively correlated with SWC, R_m, R_a, R_n, R_{an}, NO_2^--N, NO_3^--N, NH_4^+-N, MBC, MBN, MBP, TN, TC, and TP ($p < 0.01$; Fig. 5d).

Our random forest model accounted for 46.23 % of the variation in PP N_2O, identifying R_{an} and soil depth as the main driving factors that mediated the soil N_2O emissions ($p < 0.05$; Fig. S7). Structural equation modeling highlighted distinct determinants of N cycling at various stages of the sequence (Fig. 6). Throughout the chronosequence, there was a marked reduction in soil elements and microbial biomass at all depths (Fig. S3). In the natural wetlands, increasing R_n was a significant positive driver of soil PP N_2O (Fig. 4b). During reclamation, R_m played a more substantial role than Rn in affecting PP N_2O (Fig. 6b & 6c). In contrast, in the restoration phase, NO_3^--N and R_n had key positive influences on PP N_2O (Fig. 6d).

4. Discussion

4.1. Wetland reclamation and restoration altered soil N dynamics

Wetland reclamation and restoration significantly impacted biogeochemical N cycling in natural wetlands across the studied chronosequence (Fig. 2), which is consistent with previous studies (Chen et al., 2024; Wang et al., 2021). Field observations indicated that the soil N content peaked in pristine wetlands but declined over 40 years of agricultural use (Fig. 3 and Fig. S2), corroborating our initial hypothesis. This finding suggests an adaptive transition phase of the soil system in the early decades post-reclamation, which could disrupt agricultural productivity in the short term. However, prolonged cultivation has enhanced soil fertility (Aguirre-Garrido et al., 2012). Our study also revealed marked reductions in the soil TN concentration and microbial biomass N due to wetland reclamation (Fig. 2, Fig. 3, and Fig. S2), possibly due to the introduction of inadequate substrates and the diminished microbial activity necessary for N cycling (Xu et al., 2023). This reduced soil N mineralization rates during reclamation (Fig. 4).

Our findings also indicated an increase in soil TN content after 40 years of agricultural use (Fig. 3a), potentially due to altered inorganic N input from microbial processes during wetland reclamation (Liu et al., 2021). The patterns observed for MBN and inorganic N (NH_4^+-N and NO_3^--N) corroborate this trend (Fig. 3). Notably, the rate of R_m increased with the chronosequence of reclamation (Fig. 4 and Table 1), supported by the observation in subtropical cropland soils (Zhang et al., 2021). This increase could be linked to creating “microsites” within the soil matrix, a consequence of the drying-rewetting cycles characteristic of reclaimed lands (Fierer and Schimel, 2002; Gao et al., 2020; Jones et al., 2019). These microsites form a protective microaggregate pattern through the drying-rewetting process, particularly prevalent in croplands (Gupta and Roper, 2010; Jones et al., 2019). Upon the reversion of croplands to wetlands, the frequency of drying-rewetting events diminished, leading to a predictable reduction in the R_m; however, it remained lower than that in undisturbed natural wetlands (Table 1). This finding is consistent with studies on upland ecosystems, where R_m was greater in natural forests than in their restored counterparts (Liu et al., 2021). Consistent with our first hypothesis, the soil N content began to recover during the early stages of wetland restoration (3 and 5

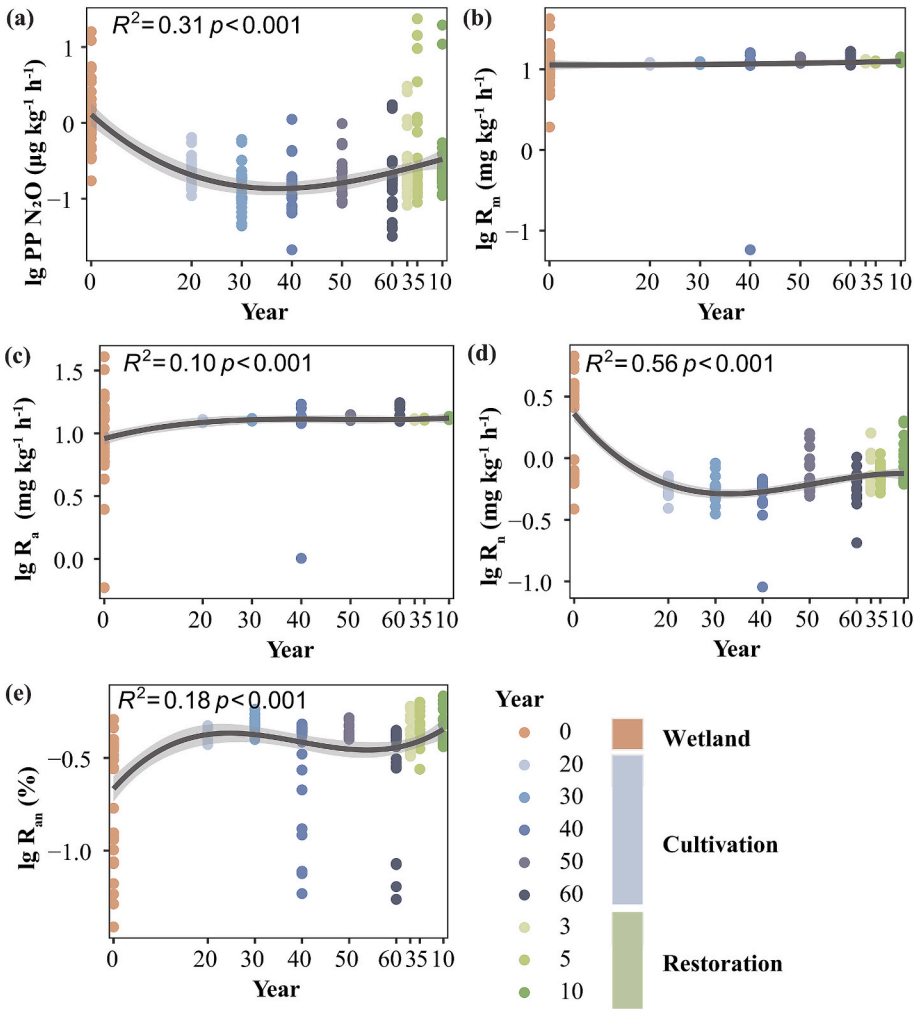


Fig. 4. Changes in soil nitrogen transformation rates the chronosequence of wetland reclamation and restoration. (a) N_2O production potential rate (PP N_2O), (b) net mineralization rate (R_m), (c) net ammonification rate (R_a), (d) net nitrification rate (R_n), and (e) nitrate in mineral N in the incubated sample (R_{an}) in different years of reclamation and restoration. Polynomial fitted curves are given, and the bands represent 95% confidence intervals. The color in each point represents the different land-use types.

years, as shown in Fig. 3). For example, Zou et al. (2011) reported that achieving near equilibrium in paddy soil N content may require more than 700 years, suggesting that a 10-year restoration period is insufficient for soils to reach an equilibrium state regarding their TN content (Fig. 3a).

Compared with natural wetland conditions, wetland reclamation led to a reduction in the rate of N transformation, while subsequent restoration facilitated faster N transformation relative to that in croplands (Fig. 4). The rate of R_n exhibited a pattern of initial decline followed by an increase throughout the reclamation and restoration sequence (Fig. 4). This trend in soil nitrification could be attributed to several factors, including soil conditions and microbial gene abundance, consistent with previous studies (Chen et al., 2020; Tang et al., 2019). Since nitrification is a critical process that consumes ammonium in soils, its rate was observed to parallel the NH_4^+-N content trends identified in our study (Fig. 3 and Fig. 4). Furthermore, a decrease in NO_3^- was observed earlier than in NH_4^+ in the early stages of reclamation (Fig. 3e), possibly due to enhanced NO_3^- leaching risks caused by tilling practices (Helfrich et al., 2020). Additionally, shifts in land use and fertilization practices were noted to diminish the abundance of nitrification genes, including *nirK*, *nirS*, and *nosZ* (Wu et al., 2017). As reclamation lands were reverted to wetland conditions, both R_n and the contents of mineral N (NH_4^+-N and NO_3^--N) began to show signs of recovery (Fig. 3c, Fig. 3d, and Fig. 4d), likely due to increased soil mineralization during the restoration phase (Ehrenfeld and Yu, 2012). The resurgence of autotrophic nitrifiers, fueled by mineral N, facilitated the oxidation of organic N and NH_4^+ to NO_3^- (Zhang et al., 2021b). This finding is consistent with the more marked recovery of NO_3^- than that of NH_4^+ observed in our study (Fig. 3).

4.2. Relationships between soil N transformation and environmental factors

Our study demonstrated that N dynamics were substantially affected by environmental variables – such as soil temperature, pH, water content, and microbial biomass – across reclamation and restoration (Fig. 5 and Fig. S6), consistent with the previous studies on the factors influencing N dynamics (Araújo et al., 2013; Banning et al., 2008). One notable observation was the disturbance of the natural wetland equilibrium due to reclamation activities (Sleeter et al., 2018). For instance, we observed a strong negative correlation between soil pH and various soil parameters, including soil inorganic N, microbial biomass, and total nutrient content throughout the reclamation and restoration sequence (Fig. 5). Although this correlation was not detected in undisturbed wetlands, it became pronounced during and after reclamation (Fig. 5b & 5c). This is likely because soil pH modulates nutrient availability by influencing microbial populations and their activities during the reclamation and restoration (Kemmitt et al., 2005). Moreover, the correlation

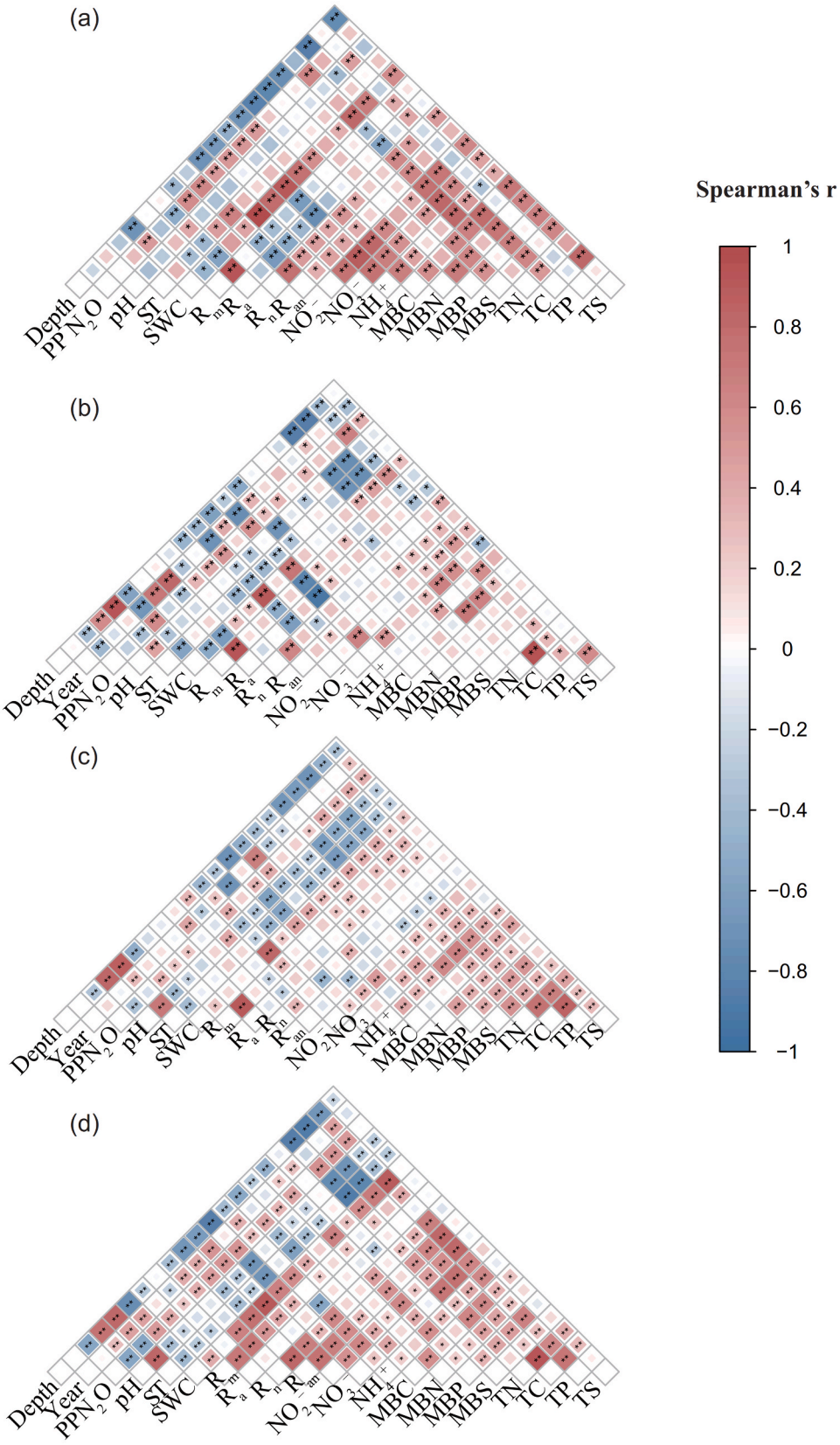


Fig. 5. Pearson correlation matrix heatmap representing geoenvironmental factors, nitrogen dynamics, and microbial biomass in natural wetlands (a), 20–30 years of wetland reclamation (b), 40–60 years of wetland reclamation (c), and restored wetlands (d). The color of the box is proportional to the Pearson's r. Significance levels are denoted with *: $p < 0.05$; **: $p < 0.01$; and ***: $p < 0.001$ in the cells. R_m, R_a, R_n, R_{an}, and PP N_2O indicate the net mineralization rates, net ammonification rates, net nitrification rates, nitrate in mineral N in the incubated sample, and N_2O production potential, respectively. Year, Depth, TC, TN, TP, TS, MBC, MBN, MBP, MBS, ST, SWC, and pH represent the year of reclamation or restoration, soil depth, soil total carbon concentration, soil total nitrogen concentration, microbial biomass carbon concentration, microbial biomass nitrogen concentration, soil temperature, soil water content, and soil pH, respectively.

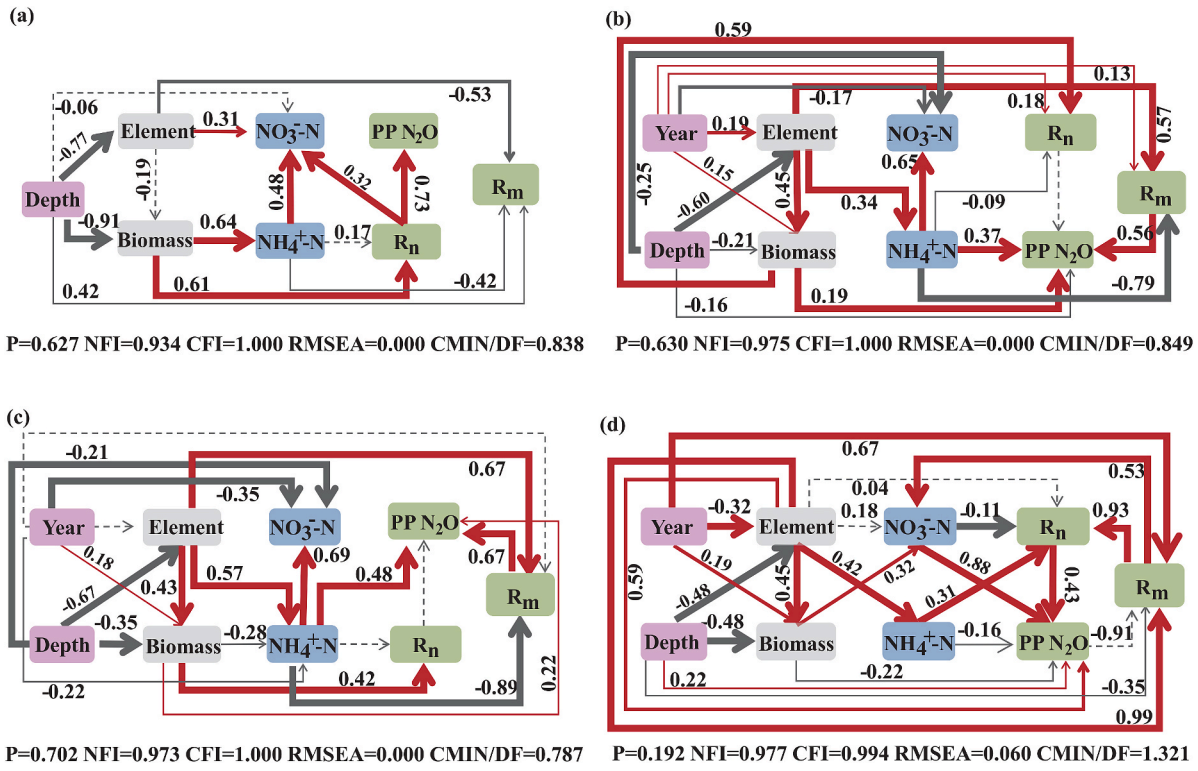


Fig. 6. Structural equation modeling for the relative controls of soil N dynamics in natural wetlands (a), 20–30 years of wetland reclamation (b), 40–60 years of wetland reclamation (c), and restored wetlands (d). Arrows in red and gray indicate positive and negative effects, respectively. The numbers adjacent to the arrows are standardized path coefficients proportional to the line thickness. Continuous and dashed arrows indicate significant and nonsignificant correlations, respectively. Arrows indicate the hypothesized direction of causation. The background color in each box represents different environmental factors and nitrogen dynamics. CFI: Comparative fit index; RMSEA: Root mean square error of approximation; CMIN/DF: Chi-square to degrees of freedom ratio. Year indicates the year of reclamation or restoration. Depth indicates the soil depth. R_m, R_n, and PP N_2O indicate the net mineralization rates, net nitrification rates, and N_2O production potential, respectively. Elements include TC, TN, TP, and TS. Biomass includes MBC, MBN, MBP, and MBS.

between soil temperature and microbial biomass components (MBC, MBN, and MBP) exhibited a similar pattern to that of soil pH across the chronosequence (Fig. 5), reflecting the sensitivity of microbial communities to temperature fluctuations (Xiao and Liang, 2023).

In pristine wetlands, soil pH did not correlate with N_2O emissions, but a significant negative correlation was observed throughout the reclamation and restoration sequence (Fig. 5). These results are consistent with a recent study by Wu et al. (2023), which indicates that soil pH is a crucial determinant of N_2O production. When the soil pH falls below 7, N_2O tends to be the dominant product of the denitrification process (Šimek et al., 2002), and denitrification rates tend to decrease as the increase of soil pH (Cuevas et al., 2020; Neina, 2019). This pattern is substantiated by our observations in which PP N_2O inversely mirrored soil pH trends across the reclamation and restoration sequence (Fig. 5, Table 1, and Table S2). Post reclamation, PP N_2O exhibited positive correlations with other environmental variables along the sequence, including soil inorganic N, microbial biomass, and total N content (Fig. 5b & 5c). Pearson correlation analysis highlighted positive associations of N_2O with R_n, R_{an}, and TN throughout the sequence (Fig. 5). These results suggest that nutrient availability may constrain the N cycle by influencing microbial activity and proliferation, disrupting the equilibrium state of natural wetlands (Kidinda et al., 2023; Zhang et al., 2023).

The interplay between soil pH and R_m differed across the reclamation and restoration sequences (Fig. 5). In undisturbed wetlands, a significant positive correlation was found between pH and both R_m and R_a (Fig. 5a). However, this relationship was inconsistent throughout the reclamation sequence (Fig. 5b, 5c & 5d). An optimal pH is known to stabilize the interaction between clays and soil organic matter (Neina, 2019), but the shift to higher pH values during reclamation appeared to weaken this stability (Table S2), subsequently leading to increased R_m (Fig. 4b). Previous studies have shown elevated R_m in alkaline soils compared to acidic soils (Curtin et al., 1998; Khalil et al., 2005), possibly due to the relatively stable pH levels observed throughout the reclamation sequence. Additionally, soil pH is a crucial regulator of biotic and abiotic factors, including TC, TP, and microbial biomass, which is consistent with our findings (Fig. 5).

4.3. Dominant drivers of N transformation along the reclamation and restoration chronosequence

Our research revealed that the influence of environmental factors on N dynamics was primarily governed by variations in soil nutrients and microbial biomass (Fig. 6 and Fig. 7). This finding is supported by the significant positive relationships between N transformation rates and microbial C metabolism at the global scale (Dai et al., 2020). N turnover rates, such as R_n, R_{an}, and inorganic N concentrations, exhibited positive correlations with TN and TC under natural wetland conditions (Fig. 5a). However, these associations diminished during the early phases of wetland reclamation, spanning 0–30 years (Fig. 5b), and were later restored over the ensuing 40–60 years of reclamation (Fig. 5c). These trends imply that TC and TN may act as constraining factors in the dynamics of the N cycle during the reclamation process (Jiang et al., 2013; Premetilake et al., 2023; Struecker and Joergensen, 2015; Zhang et al., 2007). This is supported by Li et al. (2019), which showed that net N_{min} significantly increased with soil TN on the global scale. Given the crucial role of nutrient substrates, soil TC and TN are pivotal in driving microbial diversity and activity (Fierer et al., 2003). Bannert et al. (2011) reported that the abundance of bacterial populations involved in inorganic N processing significantly decreased across a reclamation

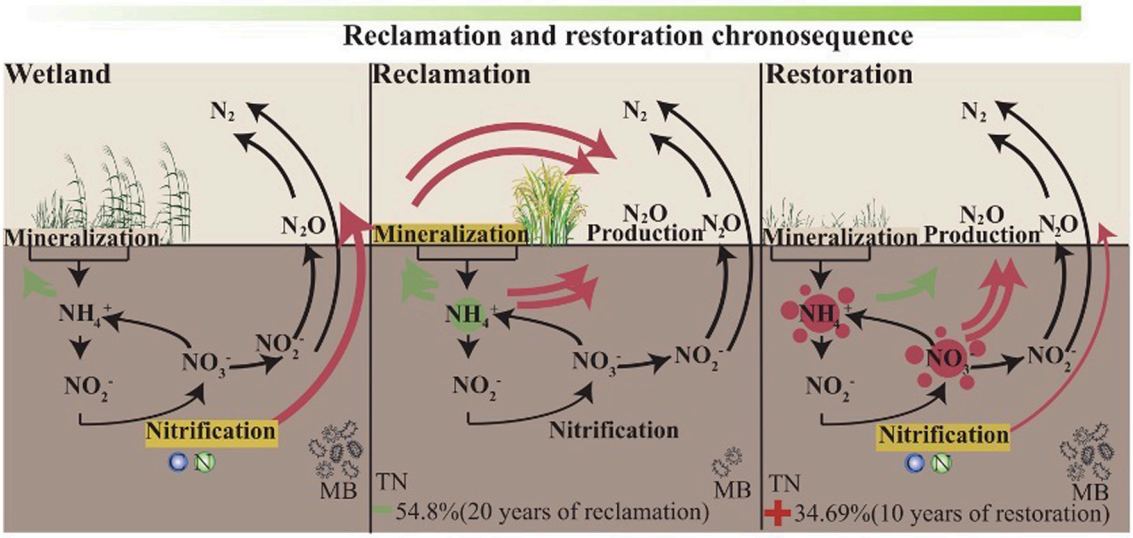


Fig. 7. An integrated perspective to understanding the shifting N cycling and soil nitrous oxide (N_2O) production potential along reclamation and restoration chronosequences in wetlands. Red arrows indicate the predominant process on N_2O production potential. Yellow boxes indicate the predominant N process on N_2O production. Green circles indicate the decreasing trends of N contents. Green arrows indicate negative effects and red arrows indicate positive effects. The net nitrification rate dominates the N_2O production potential in wetlands; the net mineralization rate and NH_4^+-N dominate the N_2O production potential in reclaimed wetlands; and the net nitrification rate, NH_4^+-N, and NO_3^--N dominate the N_2O production potential in restored wetlands. TN indicates soil total nitrogen concentration. MB indicate microbial biomass.

chronosequence. Consequently, the observed recovery of the correlation between microbial activity and N turnover rates, such as R_m, R_a, and R_{an}, was anticipated in wetlands restored from croplands (Fig. 5b).

Our random forest analysis found significant contributions of soil depth on N_2O production (Fig. S7). The rate of PP N_2O diminished with increasing soil depth (Fig. S5); this could be induced by the depth independent of environmental factors related to soil N microbial communities (Luo et al., 2021). As revealed by our results, the limitation of environmental factors on N_2O production was found more in surface than in deep soil (Fig. S8). In addition, nitrification was a pivotal factor affecting N_2O production (Fig. 6 and Fig. 7), further limiting denitrification in natural and restored wetlands (Fig. 7). We also underscored R_{an} as the most significant variable affecting PP N_2O throughout the reclamation and restoration chronosequence (Fig. S7), aligning with findings by Jiang et al. (2021). There was a strong positive correlation between PP N_2O and both R_{an} and R_n across natural wetlands, croplands, and restored wetlands (Fig. 5), which is consistent with the findings of other studies. For instance, Zhang et al. (2023a) reported that nitrification was the predominant pathway for N_2O production in alkaline Chinese soils, contributing to 81 % of N_2O emissions. Nitrification is known to supply substrates for denitrification and is widely recognized as a major source of N_2O emissions (Xu et al., 2023). Audet et al. (2014) and Liang et al. (2022), further demonstrated nitrification's significant role in N_2O emissions from wetlands and lakes, respectively.

We also found significant contributions of reclamation duration on N_2O production (Fig. S7), this could be induced by the change of environmental factors influencing soil PP N_2O along the reclamation and restoration chronosequence (Fig. 5). In the natural wetlands, R_n was not significantly correlated with ammonium or nitrate concentrations (Fig. 5a), indicating a possible saturation of the available N concentration (Bhuyan et al., 2014). Consequently, as anticipated, R_m did not influence the PP N_2O in the natural wetland (Fig. 5a). However, due to 40 to 60 years of reclamation associated with rice cultivation, PP N_2O was positively influenced by R_m (Fig. 5a & 5b), consistent with our second hypothesis. Given that soil N transformation rates were broadly linked with microbial biomass during these years of reclamation (Fig. 5c), we surmised that the increase in N_2O emissions could be modulated by N mineralization effects (Fig. 7). This is supported by previous research using a ^{15}N tracing experiment, which demonstrated that N_2O emissions could be stimulated by priming N mineralization following N fertilizer application (Xu et al., 2023). As expected, this independence between PP N_2O and R_m was seen to recover in wetlands restored from croplands (Fig. 5d).

4.4. Limitations and prospects

This study offers insights into the shifting N regime and its mechanisms across a 70-year chronosequence of wetland reclamation and restoration (Fig. 7). Three limitations were identified that deserve further investigation. First, the experimental data were derived from a single field campaign, which may not fully capture the seasonal variation in N processes. Second, while our analysis included soil R_a, R_m, and R_n rates, it is essential to recognize that gross mineralization rates could exhibit distinct patterns from those of R_m in soil (Liu et al., 2021). Detailed investigations into gross mineralization are essential to deepen our understanding of how reclamation and restoration activities influence soil N cycling. Third, while we outlined critical processes within the N cycle, our current grasp of the microbial functions that regulate N cycling remains incomplete. In summary, our assessment of N fluxes and pool sizes enriches the understanding of N dynamics and provides vital reference points for validating ecosystem models. For example, DeNitrification–DeComposition (DNDC) model, can be used to simulate the N cycle in wetland and agricultural ecosystems. The long-term evolution of N mineralization, nitrification, and denitrification can be used to optimizing the major parameters controlling the N cycling process, such as $[ON_x]$ (Concentration of NO_3^-, NO_2^-, NO and N_2O), GR_{ONx} (Maximum growth rate constant of $[ON_x]$), and M_{ONx} (Maintenance coefficient of $[ON_x]$), in DNDC or other models (Li et al., 2022; Li et al., 2023; Shen et al., 2022; Zhang et al., 2021a; Zhang et al., 2024). These insights are crucial for enhancing our understanding and management of soil N dynamics in the context of such ecological transformations.

5. Conclusion

Our study elucidates N dynamics across soil profiles and through a chronosequence of reclamation and restoration in wetlands. The

findings revealed a substantial reduction in the soil TN content by 73.19 % within the first 40 years, followed by a gradual increase during continued cultivation and restoration. The different forms of soil N exhibited distinct temporal patterns, with more significant changes in the initial 0–30 years post-reclamation than in later years. Notably, N_2O production initially decreased (0–30 years), following a slight increase during subsequent reclamation (40–60 years) and restoration phases. Soil depth and R_{an} emerged as critical variables influencing the variation in N_2O emissions, thereby influencing soil N cycling. Additionally, our results underscore the role of mineralization in influencing N cycling along a reclamation chronosequence. This study established a foundation for understanding N dynamics in response to land-use change in wetlands. An integrated approach encompassing multiple N transformation processes is crucial for assessing and managing natural wetlands. The comprehensive quantification of the N cycling response to wetland reclamation and restoration will aid in refining N cycling models. Developing N cycle process models that simulate wetland reclamation and restoration chronosequence can contribute to N loss mitigation due to land-use changes and inform future land management decisions.

CRediT authorship contribution statement

Kexin Li: Writing – original draft, Methodology, Formal analysis, Data curation. **Nannan Wang:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Rui Su:** Data curation. **Yunjiang Zuo:** Software, Formal analysis. **Jianzhao Liu:** Investigation, Data curation. **Ying Sun:** Data curation. **Xinhao Zhu:** Methodology. **Ziyu Guo:** Resources. **Lihua Zhang:** Writing – review & editing, Funding acquisition. **Changchun Song:** Project administration, Funding acquisition. **Ning Chen:** Software, Methodology. **Sergei Lupakov:** Writing – review & editing. **Fenghui Yuan:** Writing – review & editing, Supervision. **Xiaofeng Xu:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.catena.2025.108727.

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